

Milestone 2 - Project Outline

Group 6

Project Motivation and Description

Although music streaming platforms provide access to millions of songs, they often limit users to localized charts and recommendations. As a result, it's difficult for listeners, artists, and researchers to easily compare music trends across countries and genres in one place.

Our project addresses this gap by creating an interactive website that consolidates the Top 50 songs from multiple countries and genres. The platform allows users to explore and compare global music data to discover international artists and see how musical tastes vary worldwide. By centralizing and visualizing this information, our application makes it easier for users to understand global listening patterns and discover new music beyond their region.

Mandatory Application Features

- List of Top 50 Songs per Country: Users can view ranked lists of the most popular songs globally or by individual countries. Each chart entry has details such as song title, artist, genre, and chart position.
- Song/Artist/Country/Genre Filtering: Allows users to dynamically filter results by artist, country, and genre to make it easier to explore specific performers, regions, and styles.

Potential Application Features

- Music Recommendations: Personalize music recommendations based on user profile that lists artist and genre preferences.
- Interactive Visual Map: Users can hover over a country to see its top 50 songs, most popular genres and trending artists.
- Song Playback: Each chart entry includes a link or embedded player to listen directly through Spotify or Youtube.
- Lyrics Integration: Users can view lyrics for each chart entry.
- User Profiles: Registered users can save music preferences and view personalized dashboards of their top genres and artists.

Application Pages

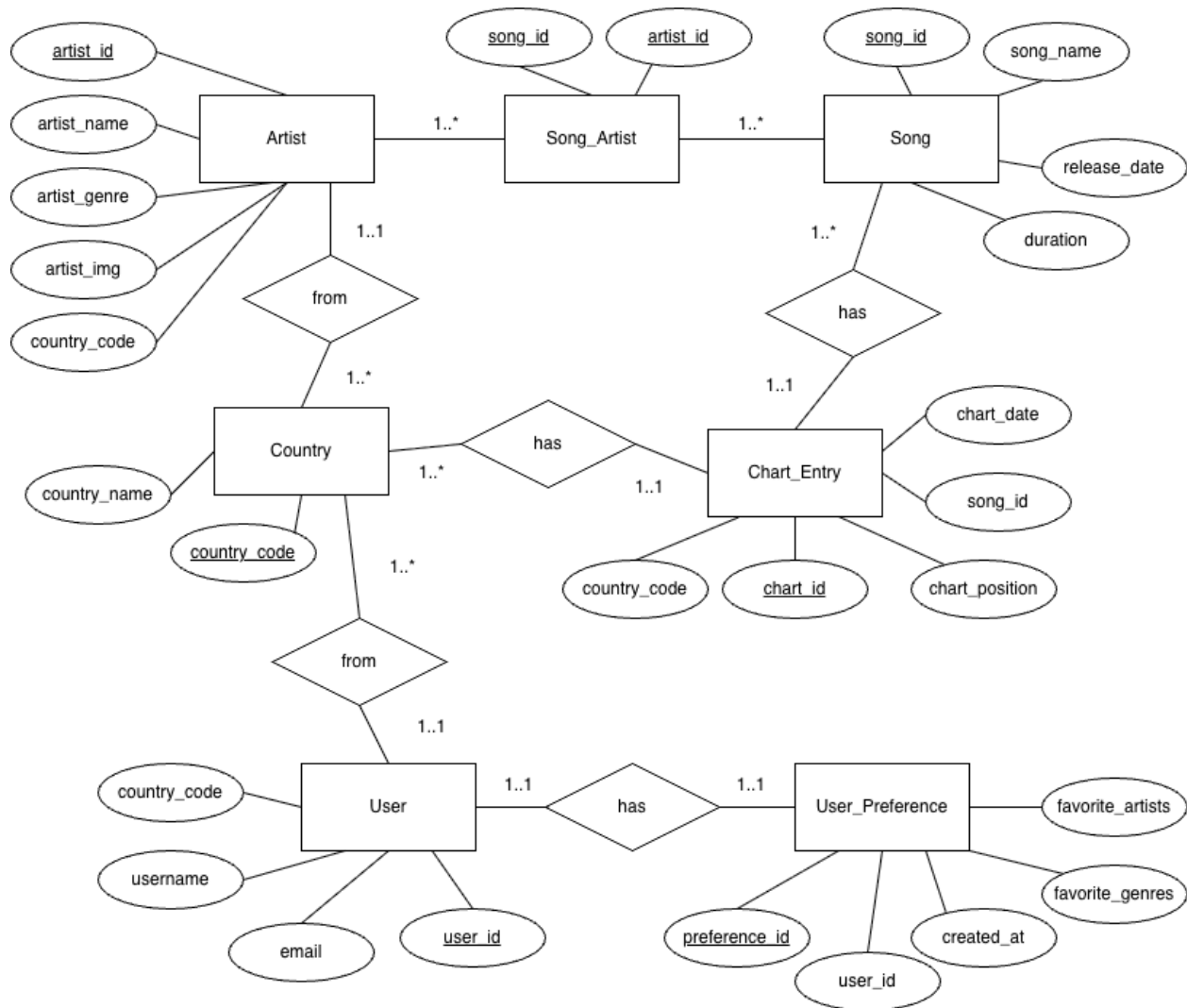
The layout for our website is to have the landing page start with a worldwide ranking of songs. On that page, users can then go to separate pages that rank the music according to Artists, or Albums. By separating the charts into songs, artists, and albums it will provide users with different perspectives to discover new music that they will like. Also, there will be a dropdown menu that allows users to filter the top charts for different countries.

As the website is under development, we hope to add additional pages for users, where they can edit their profiles and receive recommendations for songs based on the songs they like from the top charts.

A link to view the initial design of the website can be viewed here:

<https://www.figma.com/make/wsXUpykiGzHjMBU9OdV7gw/Music-Chart-Website--View-only-?node-id=0-1&t=N4bAMitOq2hDzugW-1>

Relational Schema



SQL DDL - Amy

```

CREATE TABLE country (
    country_code TEXT PRIMARY KEY,
    country_name TEXT NOT NULL
);
    
```

```

CREATE TABLE artist (
    artist_id TEXT PRIMARY KEY,
    artist_name TEXT NOT NULL,
    artist_genre TEXT,
    artist_img TEXT,
    country_code TEXT,
    FOREIGN KEY (country_code) REFERENCES country(country_code)
);
    
```

);

```
CREATE TABLE song (  
    song_id TEXT PRIMARY KEY,  
    song_name TEXT NOT NULL,  
    release_date DATE,  
    duration INTEGER);
```

```
CREATE TABLE song_artist (  
    song_id TEXT,  
    artist_id TEXT,  
    PRIMARY KEY (song_id, artist_id),  
    FOREIGN KEY (song_id) REFERENCES song(song_id),  
    FOREIGN KEY (artist_id) REFERENCES artist(artist_id)  
);
```

```
CREATE TABLE chart_entry (  
    chart_id SERIAL PRIMARY KEY,  
    song_id TEXT,  
    chart_date DATE,  
    chart_position INTEGER,  
    country_code TEXT,  
    FOREIGN KEY (song_id) REFERENCES song(song_id),  
    FOREIGN KEY (country_code) REFERENCES country(country_code)  
);
```

```
CREATE TABLE users (  
    user_id SERIAL PRIMARY KEY,  
    username TEXT NOT NULL,  
    email TEXT,  
    country_code TEXT,  
    FOREIGN KEY (country_code) REFERENCES country(country_code)  
);
```

```
CREATE TABLE user_preference (  
    preference_id SERIAL PRIMARY KEY,  
    user_id INTEGER,  
    created_at DATE,  
    favorite_genres TEXT,  
    favorite_artists TEXT,  
    FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

Data Cleaning and Pre-processing Steps

- Artists Table:
 - Split Genre table into their own rows
- Top 50 Songs Table:
 - Separate Multiple artists into their own rows
 - Add artist ID column to join onto Artists Table
 - Find Spotify Song IDs
 - Join each country's csv into one table and add country column
 - Sort by popularity
- Song ID to Artist ID
 - This will require going through the Spotify API to populate using song names and our existing Artist IDs
 - This table will allow us to join artists to their songs in the billboard

List of Technologies

- PostgreSQL
- AWS
- Spotify API
- Kaggle Datasets
- NodeJS
- ReactJS
- HTML
- CSS
- Figma

Group Member Responsibilities

- Namitra
 - Front-end development: page design and layout
 - Data pre-processing: cleaning and normalizing datasets
 - Creating individual country pages and integrating visualizations
- Kevin
 - Back-end development: setting up server, database, and API connections
 - Managing integration between datasets (Spotify API + Kaggle data)
- Jenny
 - Database design: ER diagram and relational schema
 - Data modeling: proper normalization and table relationships
- Amy
 - Implementing queries for top songs, artist rankings, and filters
 - Data analytics and visualization: derive insights from charts and trends